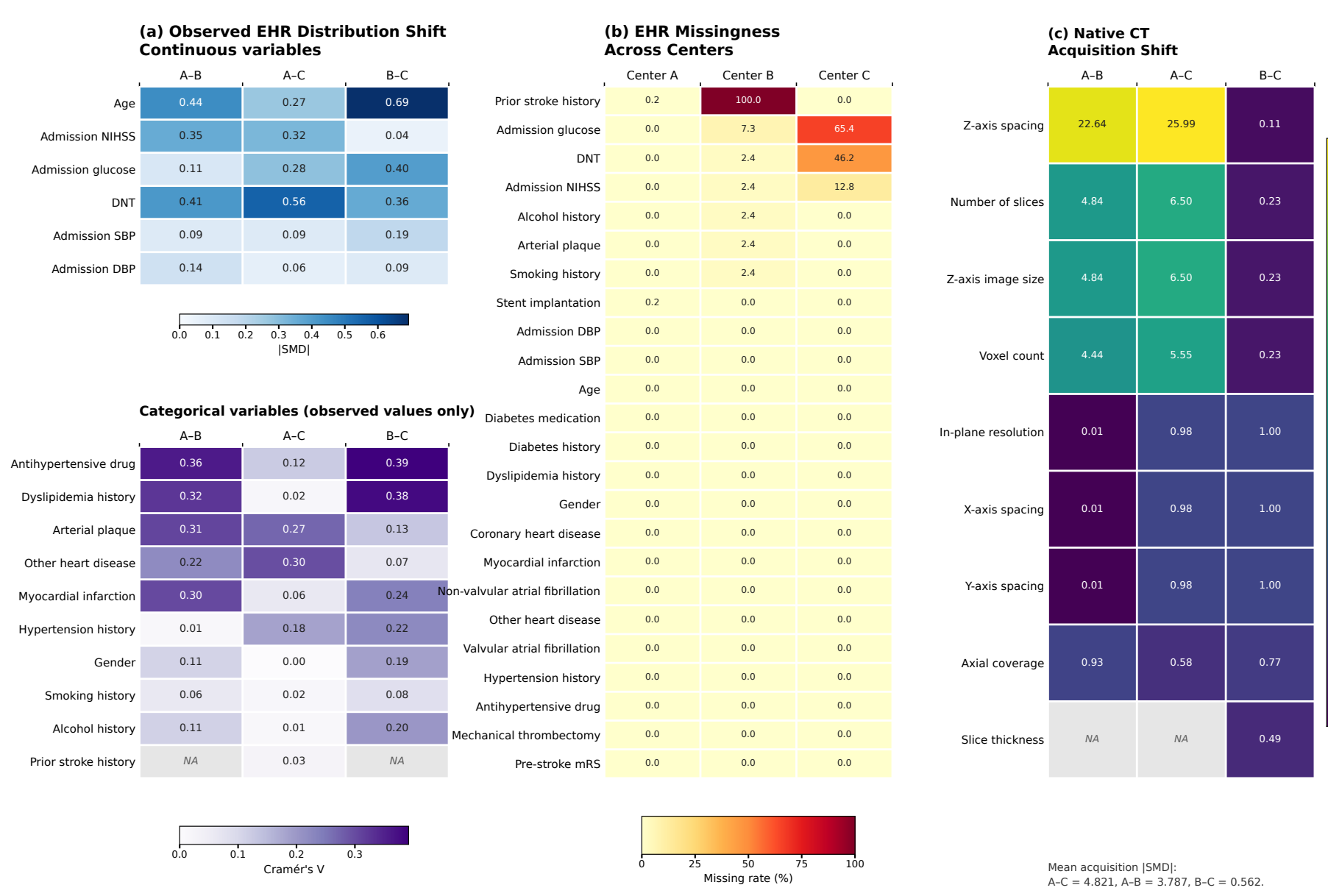

(a) Observed EHR Distribution Shift
Continuous variables
A–B
A–C
B–C
Age 0.44 0.27 0.69
Admission NIHSS 0.35 0.32 0.04
Admission glucose 0.11 0.28 0.40
DNT 0.41 0.56 0.36
Admission SBP 0.09 0.09 0.19
Admission DBP 0.14 0.06 0.09
0.0 0.1 0.2 0.3 0.4 0.5 0.6
|SMD|
Categorical variables (observed values only)
A–B
A–C
B–C
Antihypertensive drug 0.36 0.12 0.39
Dyslipidemia history 0.32 0.02 0.38
Arterial plaque 0.31 0.27 0.13
Other heart disease 0.22 0.30 0.07
Myocardial infarction 0.30 0.06 0.24
Hypertension history 0.01 0.18 0.22
Gender 0.11 0.00 0.19
Smoking history 0.06 0.02 0.08
Alcohol history 0.11 0.01 0.20
Prior stroke history NA 0.03 NA
0.0 0.1 0.2 0.3
Cramér's V
(b) EHR Missingness
Across Centers
Center A
Center B
Center C
Prior stroke history 0.2 100.0 0.0
Admission glucose 0.0 7.3 65.4
DNT 0.0 2.4 46.2
Admission NIHSS 0.0 2.4 12.8
Alcohol history 0.0 2.4 0.0
Arterial plaque 0.0 2.4 0.0
Smoking history 0.0 2.4 0.0
Stent implantation 0.2 0.0 0.0
Admission DBP 0.0 0.0 0.0
Admission SBP 0.0 0.0 0.0
Age 0.0 0.0 0.0
Diabetes medication 0.0 0.0 0.0
Diabetes history 0.0 0.0 0.0
Dyslipidemia history 0.0 0.0 0.0
Gender 0.0 0.0 0.0
Coronary heart disease 0.0 0.0 0.0
Myocardial infarction 0.0 0.0 0.0
Non-valvular atrial fibrillation 0.0 0.0 0.0
Other heart disease 0.0 0.0 0.0
Valvular atrial fibrillation 0.0 0.0 0.0
Hypertension history 0.0 0.0 0.0
Antihypertensive drug 0.0 0.0 0.0
Mechanical thrombectomy 0.0 0.0 0.0
Pre-stroke mRS 0.0 0.0 0.0
0 25 50 75 100
Missing rate (%)
(c) Native CT
Acquisition Shift
A–B
A–C
B–C
Z-axis spacing 22.64 25.99 0.11
Number of slices 4.84 6.50 0.23
Z-axis image size 4.84 6.50 0.23
Voxel count 4.44 5.55 0.23
In-plane resolution 0.01 0.98 1.00
X-axis spacing 0.01 0.98 1.00
Y-axis spacing 0.01 0.98 1.00
Axial coverage 0.93 0.58 0.77
Slice thickness NA NA 0.49
Mean acquisition |SMD|:
A–C = 4.821, A–B = 3.787, B–C = 0.562.